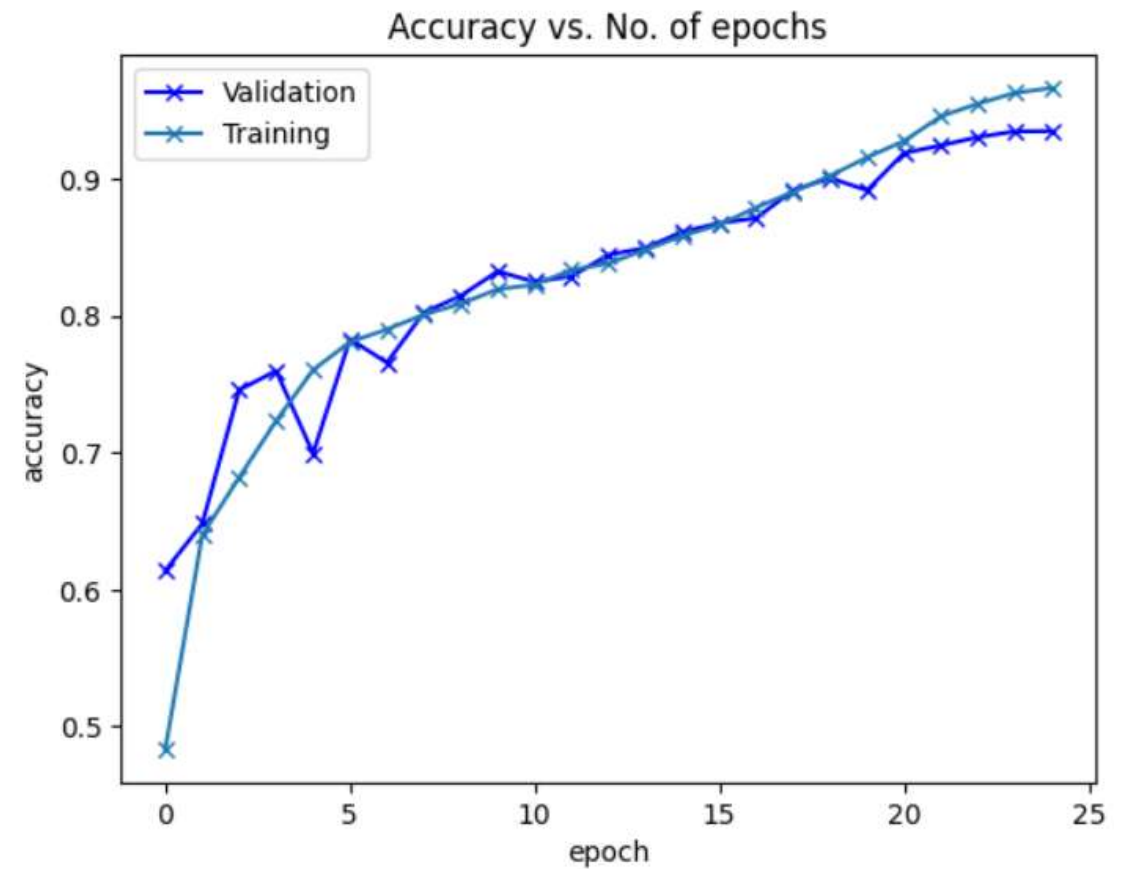
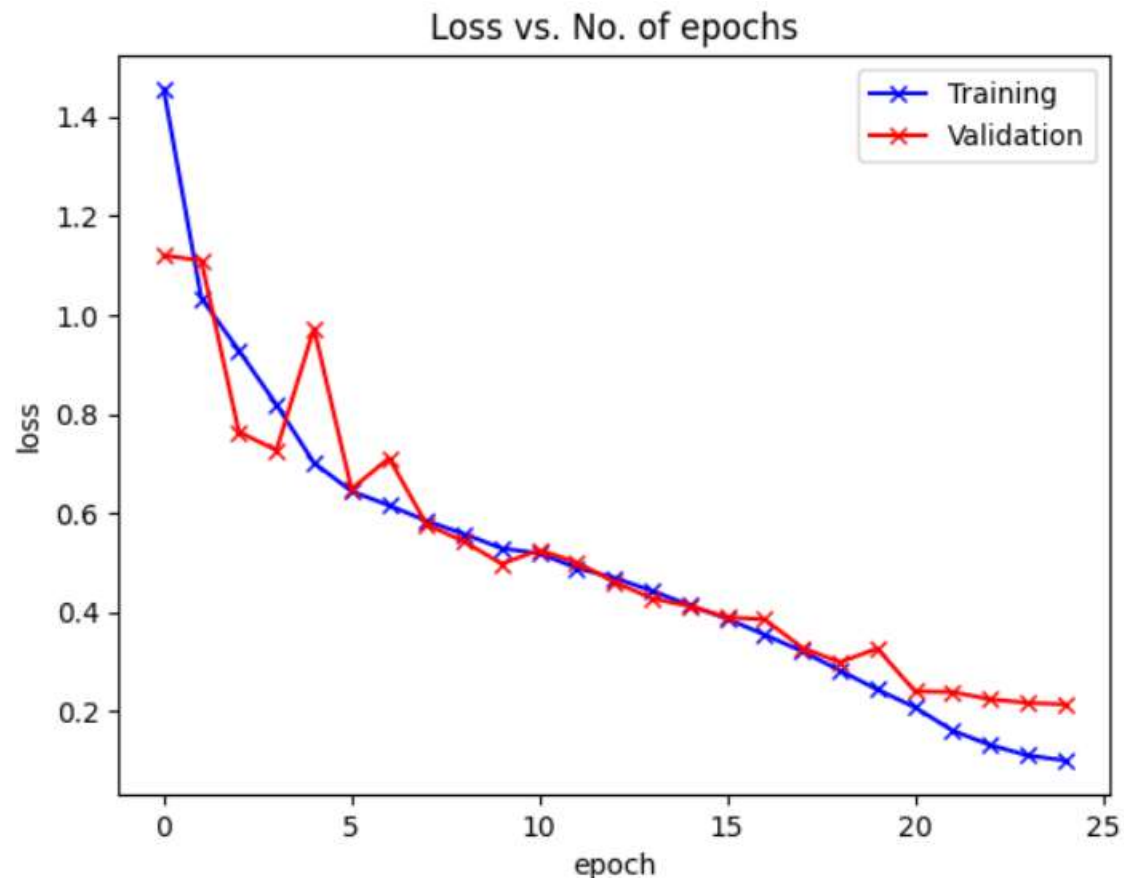
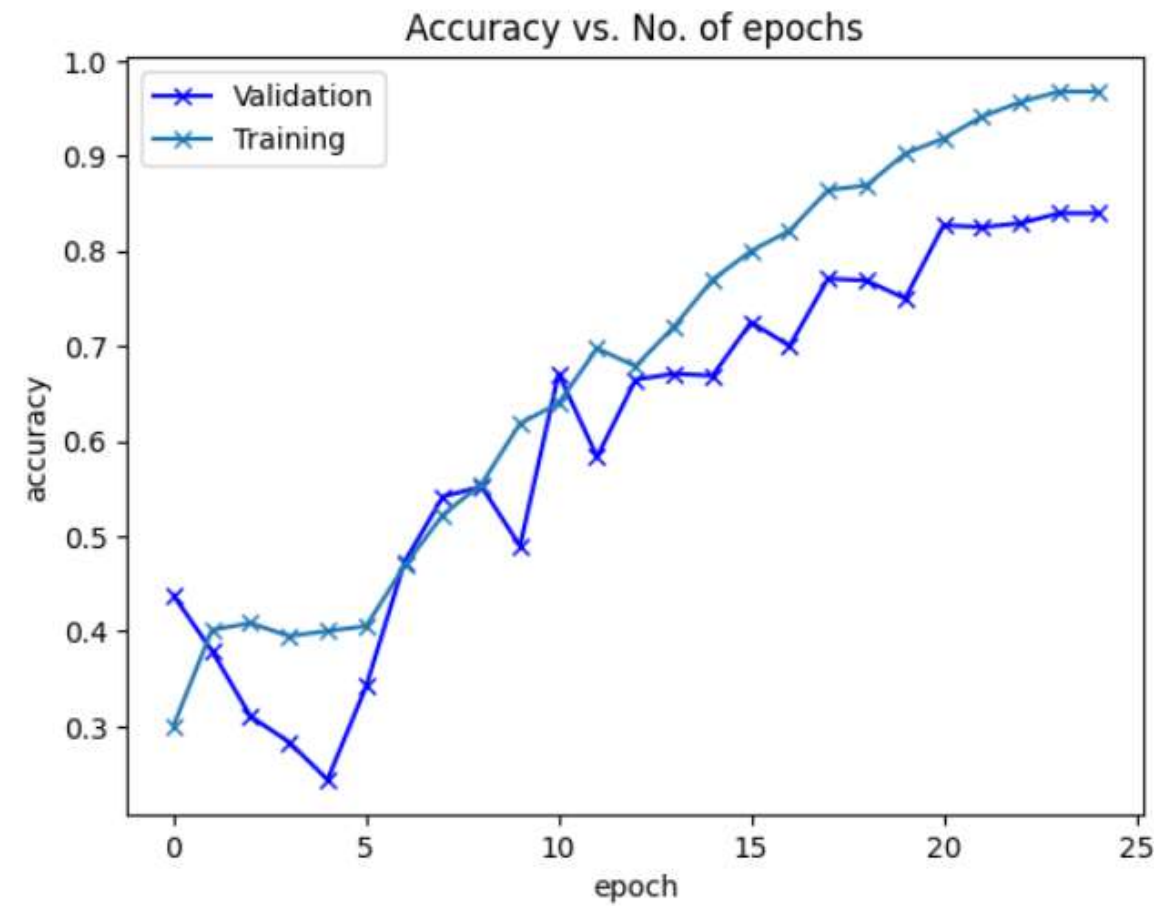
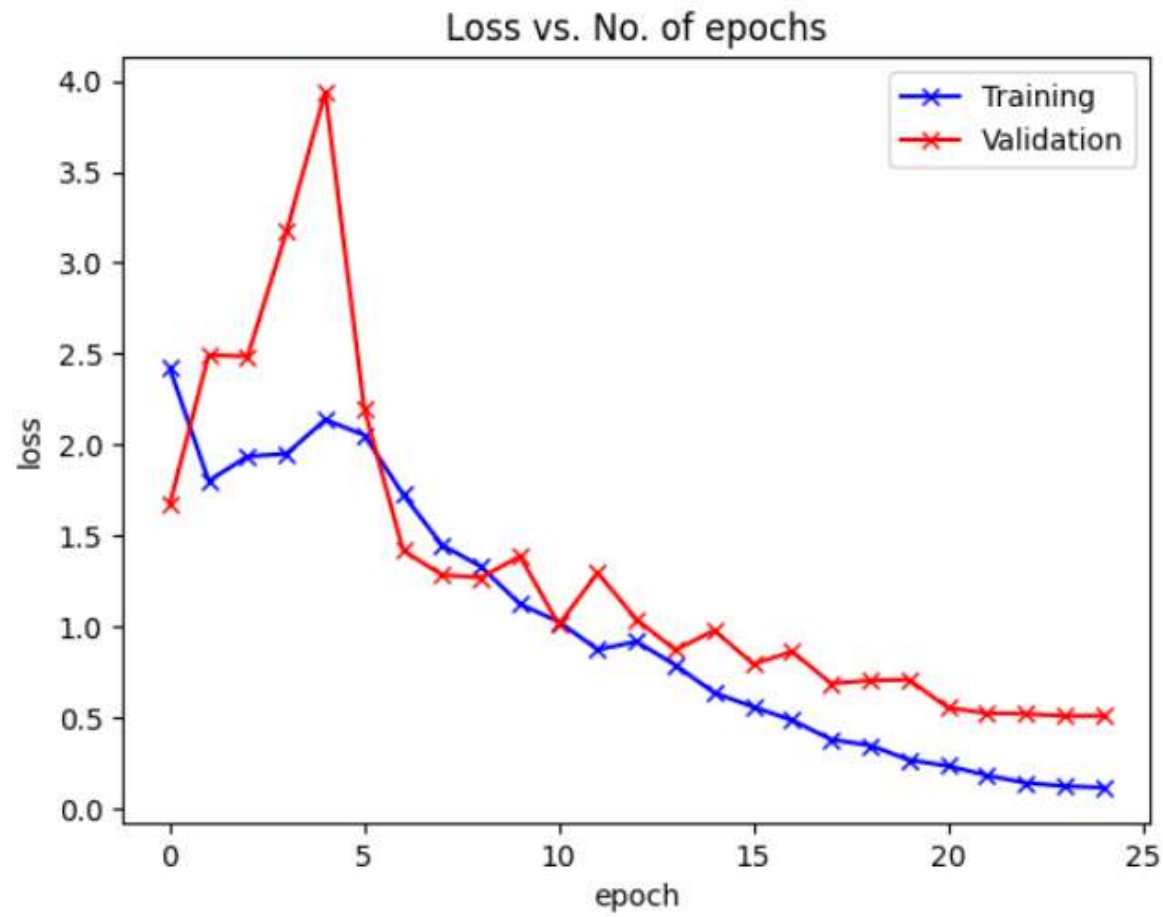


RESNET-9 ARCHITECTURE PERFORMANCE ON VARIOUS SETTINGS OF CIFAR-10, IMAGENET-256X256 DATASET

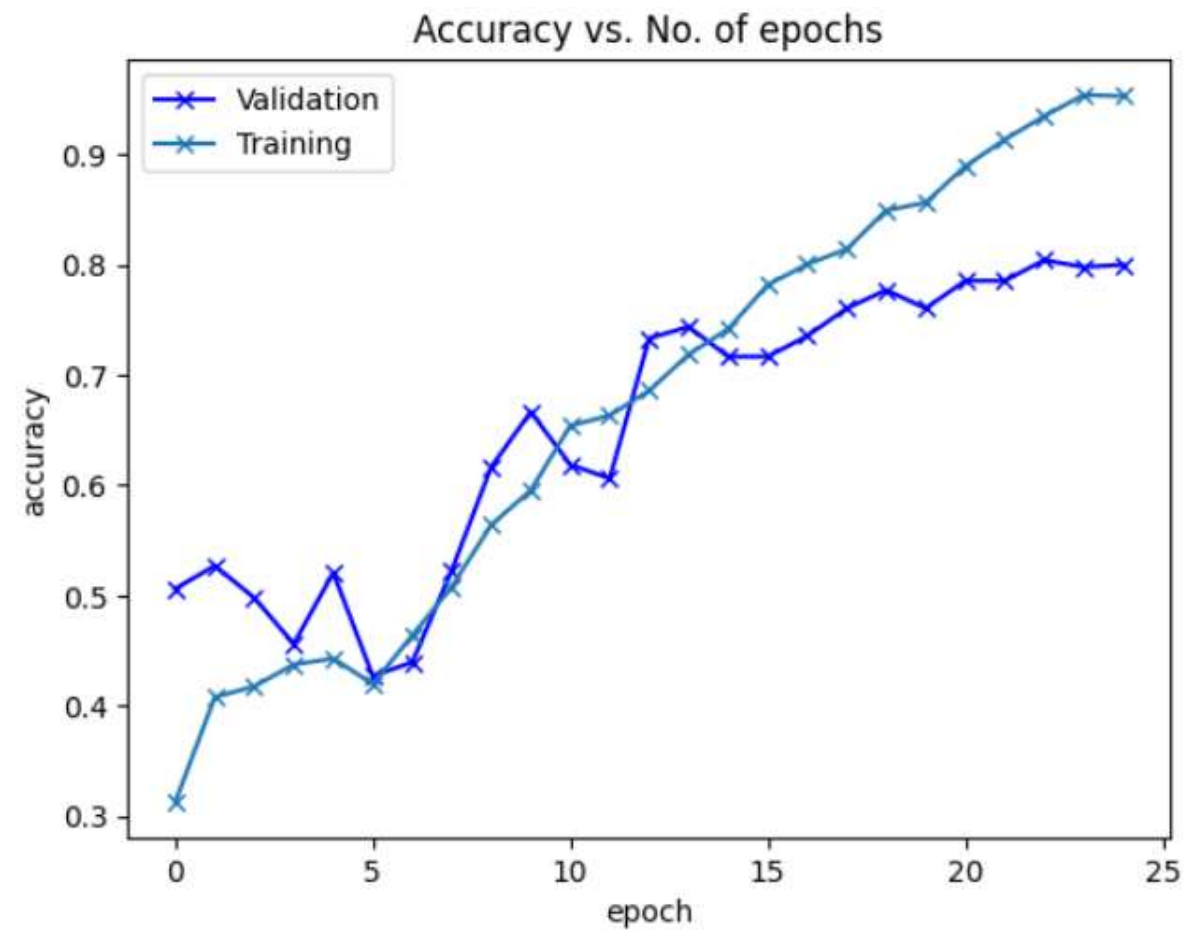
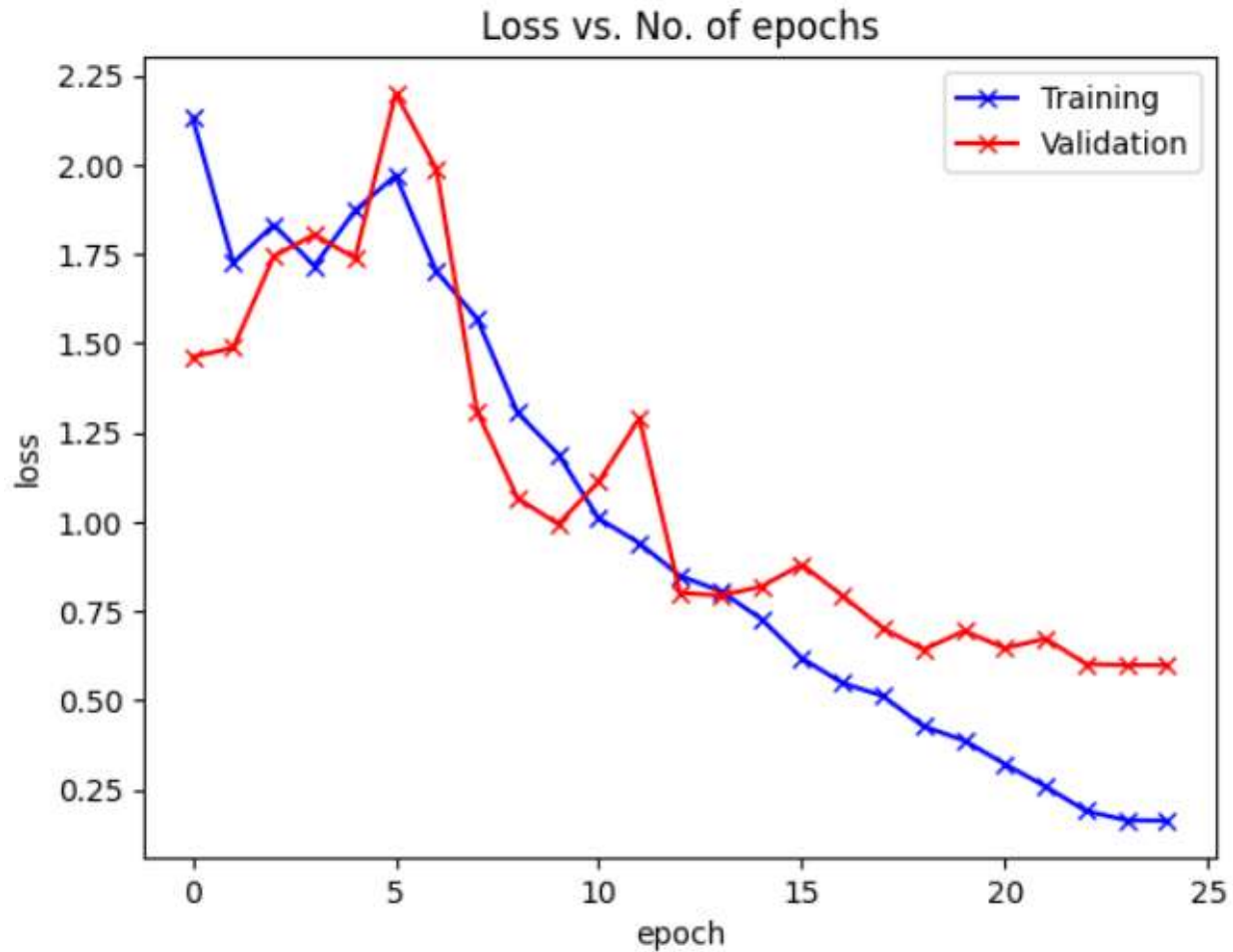
The hardware implementation of convolution module in Verilog can be found: [here](#)



Resnet9 on cifar 10 data: [colab notebook](#)



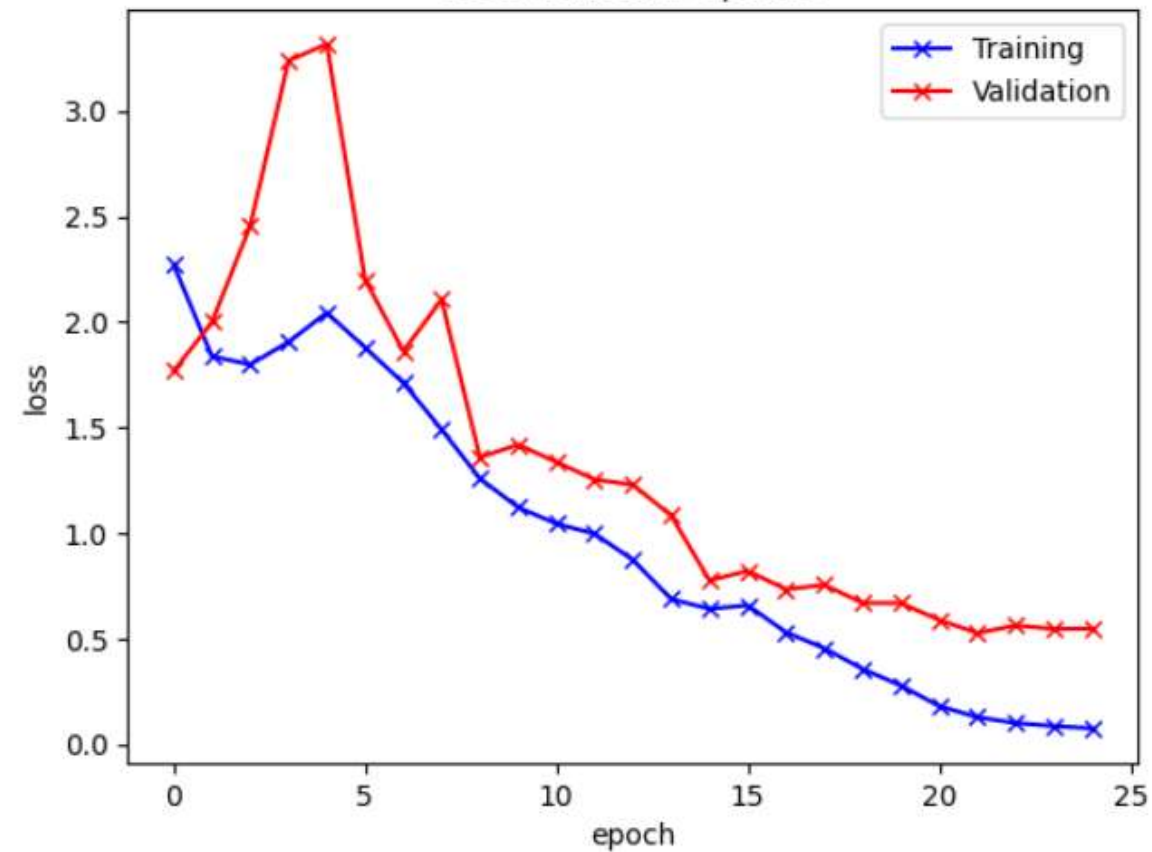
Resnet9 on original 256x256 data: [kaggle notebook](#)



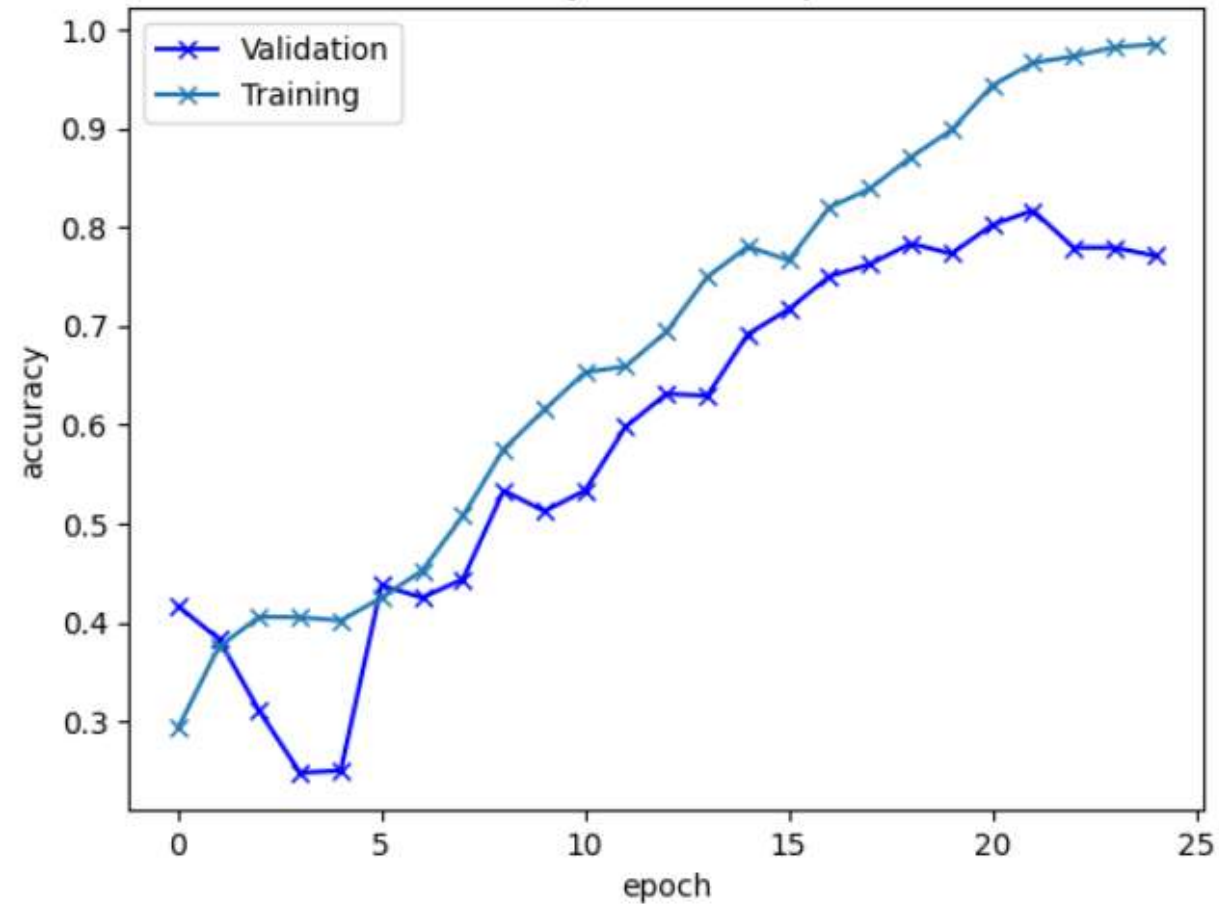
Resnet9 on Halftoned imagenet 256x256 data: [colab notebook](#)

Halftoning the imagenet 256x256 data: [kaggle notebook](#)

Loss vs. No. of epochs



Accuracy vs. No. of epochs



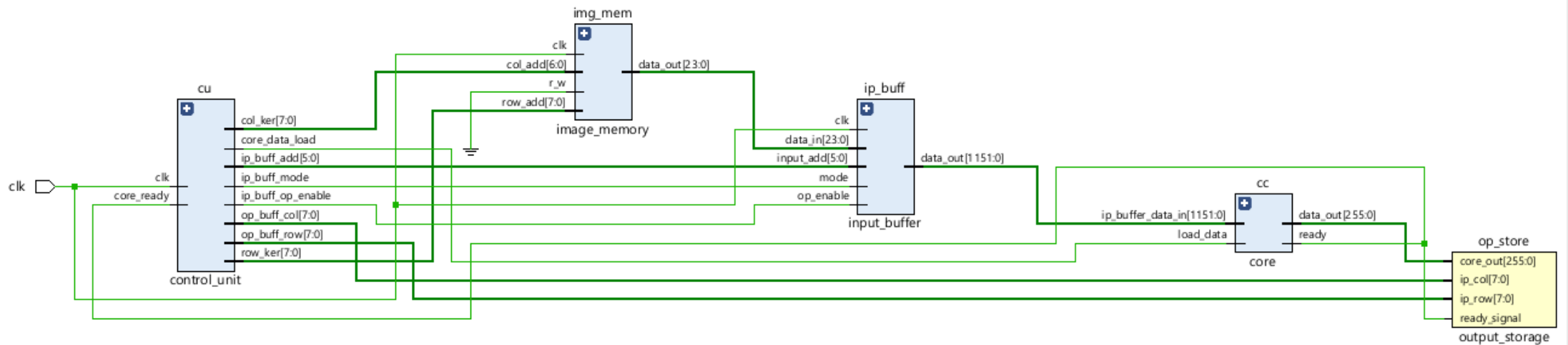
Resnet9 on dithered imagenet 256x256 data: [colab notebook](#)

dithering the imagenet 256x256 data: [kaggle notebook](#)

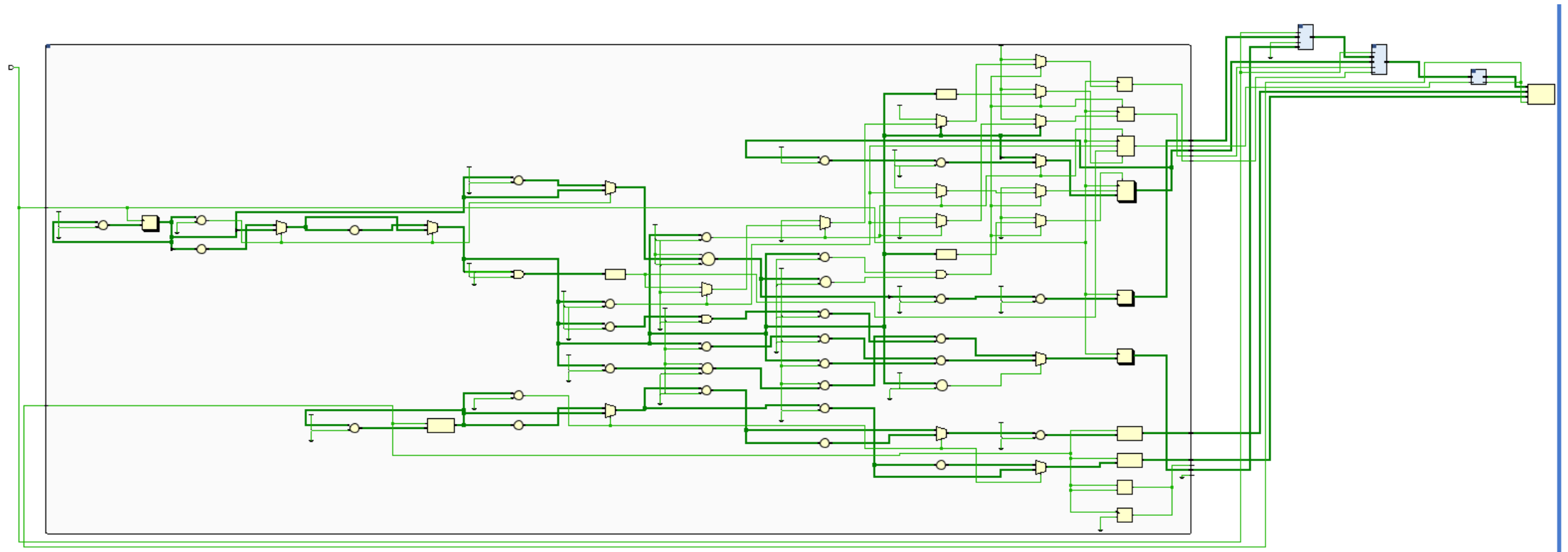
Inference (apparent)

- The model itself is trained well on the cifar-10 data but not that good on the large image data (maybe due to smaller dataset for the latter due to great training time, just 1800 images (train) and 450 images (testing)). Another practical reason maybe due to the complexity of the image compared to the simple resnet architecture (the training provides image augmentation, normalization of image pixel, proper learning rate scheduling and momentum).
- From the training on dithered and halftoned data it seems, for the halftone the training is difficult (from training loss), compared to the dithering one. The initial rise in the validation is common in both cases as the original training shows the same.
- We chose to halftone and dither on larger images compared to the cifar (32x32) as these technique outputs are visually more meaningful for larger dimensional images.

The whole convolution module schematic in verilog



The control system schematic



The shuffle buffer unit

